

Summations and Products

CS1010 Discrete Mathematics for Computer Science

Shibam Ghosh

August 20, 2026

Department of CSE, IIT Hyderabad



Summations and Products

Summations and Products

Notation

To represent the sum of terms $a_m, a_{m+1}, \dots, a_n$:

$$\sum_{j=m}^n a_j = a_m + a_{m+1} + \dots + a_n.$$

To represent the product of terms $a_m, a_{m+1}, \dots, a_n$:

$$\prod_{j=m}^n a_j = a_m \cdot a_{m+1} \cdots a_n$$

- ▶ m is the **lower limit (start)** and n is the **upper limit (end)**
- ▶ We can also sum (or product) over elements belonging to a set S : $\sum_{a \in S} f(a)$

Summing Geometric Series

Q. Let r be a real number, $r \neq 1$. Then show that $\sum_{j=0}^n ar^j = \frac{a(r^{n+1}-1)}{r-1}$

A. Proof Strategy: Let $S_n = \sum_{j=0}^n ar^j$. Multiply by r :

$$\begin{aligned} rS_n &= r \sum_{j=0}^n ar^j = \sum_{j=0}^n ar^{j+1} \\ &= \sum_{k=1}^{n+1} ar^k \quad (\text{shift index with } k = j + 1) \\ &= \left(\sum_{k=0}^n ar^k \right) + ar^{n+1} - a \\ &= S_n + ar^{n+1} - a \end{aligned}$$

$$rS_n - S_n = ar^{n+1} - a \implies S_n(r-1) = a(r^{n+1} - 1) \implies S_n = \frac{a(r^{n+1} - 1)}{r-1}$$



Summing Geometric Series

Q. Let r be a real number, $r \neq 1$. Then show that $\sum_{j=0}^n ar^j = \frac{a(r^{n+1}-1)}{r-1}$

A. Proof Strategy: Let $S_n = \sum_{j=0}^n ar^j$. Multiply by r :

$$\begin{aligned} rS_n &= r \sum_{j=0}^n ar^j = \sum_{j=0}^n ar^{j+1} \\ &= \sum_{k=1}^{n+1} ar^k \quad (\text{shift index with } k = j + 1) \\ &= \left(\sum_{k=0}^n ar^k \right) + ar^{n+1} - a \\ &= S_n + ar^{n+1} - a \end{aligned}$$

$$rS_n - S_n = ar^{n+1} - a \implies S_n(r - 1) = a(r^{n+1} - 1) \implies S_n = \frac{a(r^{n+1} - 1)}{r - 1}$$



Useful Summation Formulae

Summation	Closed-Form Formula
$\sum_{k=0}^n ar^k \ (r \neq 1)$	$\frac{a(r^{n+1}-1)}{r-1}$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k \ (x < 1)$	$\frac{1}{1-x}$

Deriving the Sum of Squares Formula

Q. Show that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

A. Consider the difference of consecutive cubes:

$$k^3 - (k-1)^3 = 3k^2 - 3k + 1 \implies 3k^2 = k^3 - (k-1)^3 + 3k - 1.$$

Now we sum this equation from $k = 1$ to n :

$$3 \sum_{k=1}^n k^2 = \sum_{k=1}^n [k^3 - (k-1)^3] + 3 \sum_{k=1}^n k - \sum_{k=1}^n 1 = n^3 + 3 \frac{n(n+1)}{2} - n.$$

Therefore,

$$\begin{aligned} \sum_{k=1}^n k^2 &= \frac{1}{3} \left(n^3 + \frac{3n(n+1)}{2} - n \right) = \frac{n(2n^2 + 3n + 1)}{6} \\ &= \boxed{\frac{n(n+1)(2n+1)}{6}}. \end{aligned}$$

Deriving the Sum of Squares Formula

Q. Show that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

A. Consider the difference of consecutive cubes:

$$k^3 - (k-1)^3 = 3k^2 - 3k + 1 \implies 3k^2 = k^3 - (k-1)^3 + 3k - 1.$$

Now we sum this equation from $k = 1$ to n ?

$$3 \sum_{k=1}^n k^2 = \sum_{k=1}^n [k^3 - (k-1)^3] + 3 \sum_{k=1}^n k - \sum_{k=1}^n 1 = n^3 + 3 \frac{n(n+1)}{2} - n.$$

Therefore,

$$\begin{aligned} \sum_{k=1}^n k^2 &= \frac{1}{3} \left(n^3 + \frac{3n(n+1)}{2} - n \right) = \frac{n(2n^2 + 3n + 1)}{6} \\ &= \boxed{\frac{n(n+1)(2n+1)}{6}}. \end{aligned}$$

Deriving the Sum of Squares Formula

Q. Show that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

A. Consider the difference of consecutive cubes:

$$k^3 - (k-1)^3 = 3k^2 - 3k + 1 \implies 3k^2 = k^3 - (k-1)^3 + 3k - 1.$$

Now we sum this equation from $k = 1$ to n :

$$3 \sum_{k=1}^n k^2 = \sum_{k=1}^n [k^3 - (k-1)^3] + 3 \sum_{k=1}^n k - \sum_{k=1}^n 1 = n^3 + 3 \frac{n(n+1)}{2} - n.$$

Therefore,

$$\begin{aligned} \sum_{k=1}^n k^2 &= \frac{1}{3} \left(n^3 + \frac{3n(n+1)}{2} - n \right) = \frac{n(2n^2 + 3n + 1)}{6} \\ &= \boxed{\frac{n(n+1)(2n+1)}{6}}. \end{aligned}$$

Deriving the Sum of Squares Formula

Q. Show that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

A. Consider the difference of consecutive cubes:

$$k^3 - (k-1)^3 = 3k^2 - 3k + 1 \implies 3k^2 = k^3 - (k-1)^3 + 3k - 1.$$

Now we sum this equation from $k = 1$ to n :

$$3 \sum_{k=1}^n k^2 = \sum_{k=1}^n [k^3 - (k-1)^3] + 3 \sum_{k=1}^n k - \sum_{k=1}^n 1 = n^3 + 3 \frac{n(n+1)}{2} - n.$$

Therefore,

$$\begin{aligned} \sum_{k=1}^n k^2 &= \frac{1}{3} \left(n^3 + \frac{3n(n+1)}{2} - n \right) = \frac{n(2n^2 + 3n + 1)}{6} \\ &= \boxed{\frac{n(n+1)(2n+1)}{6}}. \end{aligned}$$

Quick Check: Evaluating a Sum

Q. Find the value of $\sum_{j=0}^8 (1 + (-1)^j)$.

Quick Check: Evaluating a Sum

Q. Find the value of $\sum_{j=0}^8 (1 + (-1)^j)$.

A. For even j , $1 + (-1)^j = 2$, while for odd j , $1 + (-1)^j = 0$. Therefore,

$$\sum_{j=0}^8 (1 + (-1)^j) = 5(2) + 4(0) = 10$$

Quick Check: Evaluating a Sum

Q. Find the value of $\sum_{j=0}^8 (1 + (-1)^j)$.

A. For even j , $1 + (-1)^j = 2$, while for odd j , $1 + (-1)^j = 0$. Therefore,

$$\sum_{j=0}^8 (1 + (-1)^j) = 5(2) + 4(0) = 10$$

Q. Find the value of $\sum_{j=0}^8 (3^j - 2^j)$.

Quick Check: Evaluating a Sum

Q. Find the value of $\sum_{j=0}^8 (1 + (-1)^j)$.

A. For even j , $1 + (-1)^j = 2$, while for odd j , $1 + (-1)^j = 0$. Therefore,

$$\sum_{j=0}^8 (1 + (-1)^j) = 5(2) + 4(0) = 10$$

Q. Find the value of $\sum_{j=0}^8 (3^j - 2^j)$.

A. Separate the two geometric sums:

$$\sum_{j=0}^8 (3^j - 2^j) = \sum_{j=0}^8 3^j - \sum_{j=0}^8 2^j = \frac{3^9 - 1}{3 - 1} - \frac{2^9 - 1}{2 - 1} = \boxed{9330}.$$



Quick Check: Evaluating a Sum

Q. Find the value of $\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$.

Quick Check: Evaluating a Sum

Q. Find the value of $\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$.

A. Separate the sums:

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j) = 2 \sum_{j=0}^8 3^j + 3 \sum_{j=0}^8 2^j = 2 \left(\frac{3^9 - 1}{2} \right) + 3(2^9 - 1) = 21215$$

Quick Check: Evaluating a Sum

Q. Find the value of $\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$.

A. Separate the sums:

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j) = 2 \sum_{j=0}^8 3^j + 3 \sum_{j=0}^8 2^j = 2 \left(\frac{3^9 - 1}{2} \right) + 3(2^9 - 1) = 21215$$

Q. Find the value of $\sum_{j=0}^8 (2^{j+1} - 2^j)$.

Quick Check: Evaluating a Sum

Q. Find the value of $\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$.

A. Separate the sums:

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j) = 2 \sum_{j=0}^8 3^j + 3 \sum_{j=0}^8 2^j = 2 \left(\frac{3^9 - 1}{2} \right) + 3(2^9 - 1) = 21215$$

Q. Find the value of $\sum_{j=0}^8 (2^{j+1} - 2^j)$.

A. Notice that

$$2^{j+1} = 2 \cdot 2^j = 2^{j+1} - 2^j = 2^j.$$

Hence,

$$\sum_{j=0}^8 (2^{j+1} - 2^j) = \sum_{j=0}^8 2^j = \frac{2^9 - 1}{2 - 1} = 511.$$



Quick Check: Double Sums

Q. Compute $\sum_{i=0}^2 \sum_{j=0}^3 (i + j)$.

Quick Check: Double Sums

$\mathcal{Q}$. Compute $\sum_{i=0}^2 \sum_{j=0}^3 (i + j)$.

$\mathcal{A}$. For each fixed i ,

$$\sum_{j=0}^3 (i + j) = (i + 0) + (i + 1) + (i + 2) + (i + 3) = 4i + (0 + 1 + 2 + 3) = 4i + 6.$$

Therefore,

$$\begin{aligned} \sum_{i=0}^2 \sum_{j=0}^3 (i + j) &= (4(0) + 6) + (4(1) + 6) + (4(2) + 6) \\ &= 6 + 10 + 14 \\ &= \boxed{30}. \end{aligned}$$

Quick Check: Double Sums

Q. Compute $\sum_{i=0}^2 \sum_{j=1}^3 ij$.

Quick Check: Double Sums

Q. Compute $\sum_{i=0}^2 \sum_{j=1}^3 ij$.

A. Since the summand is a product of a term depending only on i and a term depending only on j , we can separate the sums:

$$\begin{aligned}\sum_{i=0}^2 \sum_{j=1}^3 ij &= \left(\sum_{i=0}^2 i \right) \left(\sum_{j=1}^3 j \right) \\ &= (0 + 1 + 2)(1 + 2 + 3) \\ &= 3 \cdot 6 \\ &= \boxed{18}.\end{aligned}$$

Quick Check: Double Sums

Q. Compute $\sum_{i=0}^2 \sum_{j=0}^3 2^i 3^j$.

Quick Check: Double Sums

Q. Compute $\sum_{i=0}^2 \sum_{j=0}^3 2^i 3^j$.

A. Since 2^i does not depend on j , we can factor it out of the inner sum:

$$\sum_{i=0}^2 2^i \sum_{j=0}^3 3^j.$$

Using the geometric sum formula,

$$\sum_{j=0}^3 3^j = 1 + 3 + 9 + 27 = 40.$$

Therefore,

$$\sum_{i=0}^2 \sum_{j=0}^3 2^i 3^j = 40 \sum_{i=0}^2 2^i = 40(1 + 2 + 4) = 40(7) = \boxed{280}.$$



Counting Elements in Sets

Comparing Cardinality

How do we compare the sizes of two sets? More importantly, how do we compare the sizes of infinite sets?

Definition

- ▶ Cantor introduced a framework to mathematically establish different sizes of infinity
- ▶ Two sets A and B have the **same cardinality**, written $|A| = |B|$, if and only if there is a bijection from A to B .

▶ If there is an injection from A to B , we write $|A| \leq |B|$.

Comparing Cardinality

How do we compare the sizes of two sets? More importantly, how do we compare the sizes of infinite sets?

Definition

- ▶ Cantor introduced a framework to mathematically establish different sizes of infinity
- ▶ Two sets A and B have the **same cardinality**, written $|A| = |B|$, if and only if there is a bijection from A to B .

- ▶ If there is an injection from A to B , we write $|A| \leq |B|$.

Countable Sets

Definition

A set S is **countable** if it is finite or has the same cardinality as the positive integers $\mathbb{Z}^+$. An infinite countable set is called **countably infinite**.

- ▶ Elements of a countable set can be listed in ordered sequence: $s_1, s_2, s_3, \dots$
- ▶ A set S is countably infinite if, there is a bijection $f : \mathbb{Z}^+ \rightarrow S$.
- ▶ The cardinal number of a countably infinite set is denoted by $\aleph_0$ ("aleph null").
- ▶ We write $|S| = \aleph_0$.

Countable Sets

Definition

A set S is **countable** if it is finite or has the same cardinality as the positive integers $\mathbb{Z}^+$. An infinite countable set is called **countably infinite**.

- ▶ Elements of a countable set can be listed in ordered sequence: $s_1, s_2, s_3, \dots$
- ▶ A set S is countably infinite if, there is a bijection $f : \mathbb{Z}^+ \rightarrow S$.
- ▶ The cardinal number of a countably infinite set is denoted by $\aleph_0$ (“aleph null”).
- ▶ We write $|S| = \aleph_0$.



Examples

Q. Show that the set of positive even integers $E = \{2, 4, 6, 8, \dots\}$ is countable.

Examples

- $\mathcal{Q}$. Show that the set of positive even integers $E = \{2, 4, 6, 8, \dots\}$ is countable.
- $\mathcal{A}$. Define the bijection $f : \mathbb{Z}^+ \rightarrow E$, $f(n) = 2n$.

Examples

- Q.* Show that the set of positive even integers $E = \{2, 4, 6, 8, \dots\}$ is countable.
- A.* Define the bijection $f : \mathbb{Z}^+ \rightarrow E$, $f(n) = 2n$.
- Q.* Show that the set of all integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is countable.

Examples

- Q.* Show that the set of positive even integers $E = \{2, 4, 6, 8, \dots\}$ is countable.
- A.* Define the bijection $f : \mathbb{Z}^+ \rightarrow E$, $f(n) = 2n$.
- Q.* Show that the set of all integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is countable.
- A.* Define the bijection $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ by:

$$f(n) = \begin{cases} (n-1)/2 & \text{if } n \text{ is odd} \\ -n/2 & \text{if } n \text{ is even} \end{cases}$$

Examples

- Q. Show that the set of positive even integers $E = \{2, 4, 6, 8, \dots\}$ is countable.
- A. Define the bijection $f : \mathbb{Z}^+ \rightarrow E$, $f(n) = 2n$.
- Q. Show that the set of all integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is countable.
- A. Define the bijection $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ by:

$$f(n) = \begin{cases} (n-1)/2 & \text{if } n \text{ is odd} \\ -n/2 & \text{if } n \text{ is even} \end{cases}$$

- *Paradoxical Twist:* An infinite set can have the same cardinality as a proper subset of itself!
- Q. Is the set of all positive integers less than 1,000,000,000 countable?



Examples

Q. Show that the set of positive even integers $E = \{2, 4, 6, 8, \dots\}$ is countable.

A. Define the bijection $f : \mathbb{Z}^+ \rightarrow E$, $f(n) = 2n$.

Q. Show that the set of all integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is countable.

A. Define the bijection $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}$ by:

$$f(n) = \begin{cases} (n-1)/2 & \text{if } n \text{ is odd} \\ -n/2 & \text{if } n \text{ is even} \end{cases}$$

► *Paradoxical Twist:* An infinite set can have the same cardinality as a proper subset of itself!

Q. Is the set of all positive integers less than 1,000,000,000 countable?

A. Yes! In fact, this is a finite set!



The Union of Two Countable Sets

Theorem

If A and B are countably infinite sets, then $A \cup B$ is countable.

We can write A and B as $A = \{a_1, a_2, a_3, \dots\}$, and $B = \{b_1, b_2, b_3, \dots\}$.

$$\underbrace{a_1}_1, \underbrace{b_1}_2, \underbrace{a_2}_3, \underbrace{b_2}_4, \underbrace{a_3}_5, \underbrace{b_3}_6, \dots$$

This gives a list containing every element of $A \cup B$. If an element occurs twice because $A \cap B \neq \emptyset$, simply **skip the duplicate**. Thus, we can list every element of $A \cup B$ using the positive integers.

The Union of Two Countable Sets

Theorem

If A and B are countably infinite sets, then $A \cup B$ is countable.

We can write A and B as $A = \{a_1, a_2, a_3, \dots\}$, and $B = \{b_1, b_2, b_3, \dots\}$.

$$\underbrace{a_1}_1, \underbrace{b_1}_2, \underbrace{a_2}_3, \underbrace{b_2}_4, \underbrace{a_3}_5, \underbrace{b_3}_6, \dots$$

This gives a list containing every element of $A \cup B$. If an element occurs twice because $A \cap B \neq \emptyset$, simply **skip the duplicate**. Thus, we can list every element of $A \cup B$ using the positive integers.

The Union of Two Countable Sets

Theorem

If A and B are countably infinite sets, then $A \cup B$ is countable.

We can write A and B as $A = \{a_1, a_2, a_3, \dots\}$, and $B = \{b_1, b_2, b_3, \dots\}$.

$$\underbrace{a_1}_1, \underbrace{b_1}_2, \underbrace{a_2}_3, \underbrace{b_2}_4, \underbrace{a_3}_5, \underbrace{b_3}_6, \dots$$

This gives a list containing every element of $A \cup B$. If an element occurs twice because $A \cap B \neq \emptyset$, simply **skip the duplicate**. Thus, we can list every element of $A \cup B$ using the positive integers.

Uncountability and Subsets

Theorem

If A and B are sets where A is uncountable and $A \subseteq B$, then B is uncountable.

Suppose, that B is countable. Then B can be listed as $B = \{b_1, b_2, b_3, \dots\}$.

Because $A \subseteq B$, take the subsequence of $\{b_n\}$ consisting of exactly those terms that belong to A . This gives a listing of all the elements of A :

$$A = \{b_{n_1}, b_{n_2}, b_{n_3}, \dots\}.$$

But this is impossible because A is uncountable.

Uncountability and Subsets

Theorem

If A and B are sets where A is uncountable and $A \subseteq B$, then B is uncountable.

Suppose, that B is countable. Then B can be listed as $B = \{b_1, b_2, b_3, \dots\}$.

Because $A \subseteq B$, take the subsequence of $\{b_n\}$ consisting of exactly those terms that belong to A . This gives a listing of all the elements of A :

$$A = \{b_{n_1}, b_{n_2}, b_{n_3}, \dots\}.$$

But this is impossible because A is uncountable.

Quick Check: Finite or Infinite?

Q. Determine whether the set $A \times \mathbb{Z}^+$ is finite, countably infinite, or uncountable, where $A = \{2, 3\}$.

Quick Check: Finite or Infinite?

Q. Determine whether the set $A \times \mathbb{Z}^+$ is finite, countably infinite, or uncountable, where $A = \{2, 3\}$.

A. Since $A \times \mathbb{Z}^+ = \{(2, 1), (2, 2), (2, 3), \dots, (3, 1), (3, 2), (3, 3), \dots\}$, the set is infinite.

Quick Check: Finite or Infinite?

Q. Determine whether the set $A \times \mathbb{Z}^+$ is finite, countably infinite, or uncountable, where $A = \{2, 3\}$.

A. Since $A \times \mathbb{Z}^+ = \{(2, 1), (2, 2), (2, 3), \dots, (3, 1), (3, 2), (3, 3), \dots\}$, the set is infinite.

A. Moreover, we can arrange its elements in a sequence:

$$(2, 1), (3, 1), (2, 2), (3, 2), (2, 3), (3, 3), \dots$$

Therefore,

$A \times \mathbb{Z}^+$ is countably infinite.
--

Quick Check: Finite or Infinite?

2. Determine whether the set of all bit strings not containing the bit 0 is finite, countably infinite, or uncountable.

Quick Check: Finite or Infinite?

Q. Determine whether the set of all bit strings not containing the bit 0 is finite, countably infinite, or uncountable.

A. If a bit string contains no 0, every bit must be 1. Thus, the set is

$$\{\epsilon, 1, 11, 111, 1111, \dots\},$$

where ϵ denotes the empty string.

A. There is exactly one such string of each length. We can therefore list them according to length. Hence,

The set is (countable + infinite) $\implies$ countably infinite.
--

Quick Check: Finite or Infinite?

- Q. Determine whether the set of integers divisible by 5 but not by 7 is finite, countably infinite, or uncountable.

Quick Check: Finite or Infinite?

- $\mathcal{Q}$. Determine whether the set of integers divisible by 5 but not by 7 is finite, countably infinite, or uncountable.
- $\mathcal{A}$. Every such integer has the form $5k$, $k \in \mathbb{Z}$, except when k is divisible by 7. For example,

$$\dots, -20, -10, 5, 10, 15, 20, 30, 40, 45, \dots$$

There are infinitely many such integers. Moreover, this set is a subset of $\mathbb{Z}$, and $\mathbb{Z}$ is countable. Therefore, the set is at most countable and, since it is infinite,

the set is countably infinite.

Quick Check: Differences of Uncountable Sets

Q. Give examples of two uncountable sets A and B such that $A - B$ is
(a) finite, (b) countably infinite, (c) uncountable.

Quick Check: Differences of Uncountable Sets

Q. Give examples of two uncountable sets A and B such that $A - B$ is
(a) finite, (b) countably infinite, (c) uncountable.

A. (a) **Finite:** Take $A = \mathbb{R}$, $B = \mathbb{R} \setminus \{0, 1\}$. Then $A - B = \{0, 1\}$, which is finite.

Quick Check: Differences of Uncountable Sets

Q. Give examples of two uncountable sets A and B such that $A - B$ is
(a) finite, (b) countably infinite, (c) uncountable.

A. (a) **Finite:** Take $A = \mathbb{R}$, $B = \mathbb{R} \setminus \{0, 1\}$. Then $A - B = \{0, 1\}$, which is finite.

A. (b) **Countably infinite:** Take $A = \mathbb{R}$, $B = \mathbb{R} \setminus \mathbb{Z}$. Then $A - B = \mathbb{Z}$, which is countably infinite.

Quick Check: Differences of Uncountable Sets

Q. Give examples of two uncountable sets A and B such that $A - B$ is
(a) finite, (b) countably infinite, (c) uncountable.

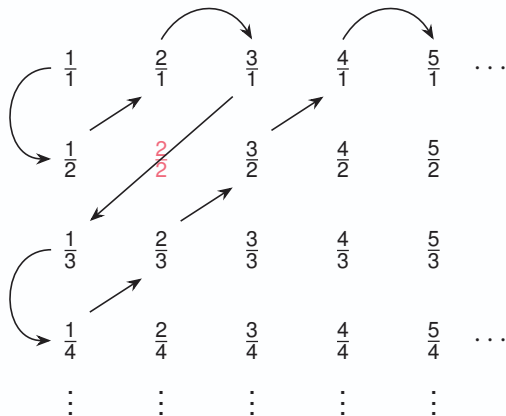
A. (a) **Finite:** Take $A = \mathbb{R}$, $B = \mathbb{R} \setminus \{0, 1\}$. Then $A - B = \{0, 1\}$, which is finite.

A. (b) **Countably infinite:** Take $A = \mathbb{R}$, $B = \mathbb{R} \setminus \mathbb{Z}$. Then $A - B = \mathbb{Z}$, which is countably infinite.

A. (c) **Uncountable:** Take $A = \mathbb{R}$, $B = (0, \infty)$. Then $A - B = (-\infty, 0]$, which is uncountable.

The Set of Positive Rational Numbers $\mathbb{Q}^+$ is Countable

- ▶ Every positive rational number is the quotient $\frac{p}{q}$ of two positive integers
- ▶ **Cantor's Grid:** Arrange positive rationals p/q in a grid where row = p and column = q .
- ▶ Traverse along diagonals.
- ▶ Skip values that have already been listed in reduced form (e.g., skip $2/2$ as it is equal to $1/1$).
- ▶ This process successfully lists all positive rational numbers in a sequence!



The set of Real Numbers $\mathbb{R}$ is Uncountable

Cantor's Diagonalization Argument: Assume there's one to one correspondence between the real numbers and the natural numbers. Consequently, we exhaustively list them as follows:

$$r_1 = 0.d_{11}d_{12}d_{13}d_{14}\dots$$

$$r_2 = 0.d_{21}d_{22}d_{23}d_{24}\dots$$

$$r_3 = 0.d_{31}d_{32}d_{33}d_{34}\dots$$

$$\vdots$$

Construct a new real number $b = 0.b_1b_2b_3b_4\dots$:

$$b_i = \begin{cases} 1 & \text{if } d_{ii} = 9 \\ 9 - d_{ii} & \text{if } d_{ii} \in \{0, \dots, 8\} \end{cases}$$

b differs from r_i at the i -th decimal place, meaning b is not in the list!

The set of Real Numbers $\mathbb{R}$ is Uncountable

Cantor's Diagonalization Argument: Assume there's one to one correspondence between the real numbers and the natural numbers. Consequently, we exhaustively list them as follows:

$$r_1 = 0.d_{11}d_{12}d_{13}d_{14}\dots$$

$$r_2 = 0.d_{21}d_{22}d_{23}d_{24}\dots$$

$$r_3 = 0.d_{31}d_{32}d_{33}d_{34}\dots$$

$$\vdots$$

Construct a new real number $b = 0.b_1b_2b_3b_4\dots$:

$$b_i = \begin{cases} 1 & \text{if } d_{ii} = 9 \\ 9 - d_{ii} & \text{if } d_{ii} \in \{0, \dots, 8\} \end{cases}$$

b differs from r_i at the i -th decimal place, meaning b is not in the list!

Matrices

What is a Matrix?

Definition

A **matrix** is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ **matrix**.

- ▶ If $m = n$, the matrix is called **square**.
- ▶ We denote elements with indices: a_{ij} is the element in row i and column j .

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Matrix Addition

Definition

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ matrices. The sum $A + B$ is the $m \times n$ matrix whose elements are $a_{ij} + b_{ij}$ for all i, j .

Matrices must have the exact same dimensions to be added.

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 3 & 4 & 0 \\ 1 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & -1 \\ 3 & 0 & 5 \end{bmatrix}$$

Matrix Multiplication

Definition

Let A be an $m \times k$ matrix and B be a $k \times n$ matrix. The product AB is the $m \times n$ matrix $C = [c_{ij}]$ where:

$$c_{ij} = \sum_{s=1}^k a_{is}b_{sj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}$$

Dimension Rule: The number of columns in A = the number of rows in B .

$$\boxed{m \times k} \times \boxed{k \times n} = \boxed{m \times n}$$

Matrix Multiplication: Example

Multiply the matrices

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 5 \\ 2 & 0 \end{bmatrix}.$$

Solution:

$$AB = \begin{bmatrix} 2 \cdot 1 + (-1) \cdot 2 & 2 \cdot 5 + (-1) \cdot 0 \\ 3 \cdot 1 + 4 \cdot 2 & 3 \cdot 5 + 4 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 10 \\ 11 & 15 \end{bmatrix}.$$

Question: Is $AB = BA$?

$$BA = \begin{bmatrix} 1 & 5 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 17 & 19 \\ 4 & -2 \end{bmatrix}. \quad \text{Hence, } \boxed{AB \neq BA}$$

Hence, matrix multiplication is generally **not commutative**.

Matrix Multiplication: Example

Multiply the matrices

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 5 \\ 2 & 0 \end{bmatrix}.$$

Solution:

$$AB = \begin{bmatrix} 2 \cdot 1 + (-1) \cdot 2 & 2 \cdot 5 + (-1) \cdot 0 \\ 3 \cdot 1 + 4 \cdot 2 & 3 \cdot 5 + 4 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 10 \\ 11 & 15 \end{bmatrix}.$$

Question: Is $AB = BA$?

$$BA = \begin{bmatrix} 1 & 5 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 17 & 19 \\ 4 & -2 \end{bmatrix}. \quad \text{Hence, } \boxed{AB \neq BA}$$

Hence, matrix multiplication is generally not commutative.

Matrix Multiplication: Example

Multiply the matrices

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 5 \\ 2 & 0 \end{bmatrix}.$$

Solution:

$$AB = \begin{bmatrix} 2 \cdot 1 + (-1) \cdot 2 & 2 \cdot 5 + (-1) \cdot 0 \\ 3 \cdot 1 + 4 \cdot 2 & 3 \cdot 5 + 4 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 10 \\ 11 & 15 \end{bmatrix}.$$

Question: Is $AB = BA$?

$$BA = \begin{bmatrix} 1 & 5 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 17 & 19 \\ 4 & -2 \end{bmatrix}. \quad \text{Hence, } \boxed{AB \neq BA}$$

Hence, matrix multiplication is generally **not commutative**.

Matrix Multiplication: Example

Multiply

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}.$$

Solution:

$$AB = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 3 & 1 \cdot 1 + 2 \cdot 0 & 1 \cdot 4 + 2 \cdot 5 \\ 0 \cdot 2 + 3 \cdot 3 & 0 \cdot 1 + 3 \cdot 0 & 0 \cdot 4 + 3 \cdot 5 \end{bmatrix} = \begin{bmatrix} 8 & 1 & 14 \\ 9 & 0 & 15 \end{bmatrix}.$$

Since A is 2×2 and B is 2×3 , BA is undefined $(2 \times 3)(2 \times 2)$,

Matrix Multiplication: Example

Multiply

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}.$$

Solution:

$$AB = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 3 & 1 \cdot 1 + 2 \cdot 0 & 1 \cdot 4 + 2 \cdot 5 \\ 0 \cdot 2 + 3 \cdot 3 & 0 \cdot 1 + 3 \cdot 0 & 0 \cdot 4 + 3 \cdot 5 \end{bmatrix} = \begin{bmatrix} 8 & 1 & 14 \\ 9 & 0 & 15 \end{bmatrix}.$$

Since A is 2×2 and B is 2×3 , BA is undefined $(2 \times 3)(2 \times 2)$,

Special Matrices

1. Zero Matrix ($0_{m \times n}$)

A matrix of any dimension where every entry is 0.

$$0_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

2. Diagonal Matrix

A square matrix ($n \times n$) where all entries outside the main diagonal are 0.

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

3. Identity Matrix (I_n)

A special diagonal matrix where all main diagonal entries are exactly 1. For any $n \times n$ matrix A , $I_n A = A I_n = A$.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Transpose of Matrices

Transpose Matrix (A^T)

The matrix obtained by swapping the rows and columns of A . If $A = [a_{ij}]$, then $A^T = [a_{ji}]$.

Transpose Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

A square matrix is called **symmetric** if $A = A^T$.



Thank You for your
attention!

Any questions?